

Research Proposal & PRD- designing

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Version Control

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1. AIBS – Research Proposal (first draft)

1.1 Target Audience

Mechanical engineering and industrial product design firms in the Netherlands, with 20–75 employees, that develop custom machinery, production equipment, or technical components using CAD software such as SolidWorks, Autodesk Inventor, or Siemens NX. In many of these companies, engineers and designers spend a significant amount of time creating concepts, modifying existing designs, generating technical drawings, and reviewing design alternatives. Their question is not which AI tool to use, but whether AI can meaningfully improve their design pipeline by reducing design time, increasing design quality, and automating repetitive engineering tasks, and whether the benefits justify the required investment and implementation effort.

1.2 What do these companies need?

We see that companies are too afraid to start using AI in their business. Because they think the way of working now is perfect for them. There is also a lack of knowledge about the things AI can do, by the people who make these decisions in these businesses. But they are forced to change so they need to decide are we going to transform our company and be a more future realistic company, or fall behind and let other rival companies capitalize in their market.

According to Draghi the innovation gap between the US/China and Europe is big so companies in Europe need to decide if they want to close that gap or not. Because it's in their own hands if companies want to be more successful according to Draghi they need to increase their investment more in AI, digital technologies, research and innovation. Instead of focussing on the present and results.

1.3 Questions & Methodology

1.3.1 Main question

How can designers and engineers use AI to improve their design pipeline?

1.3.2 Sub questions

Sub-question 1: What does a design engineer do on a daily bases?	
Sub-question type	Field
Goal	Current work for a design engineer
Research-strategy	Document analysis - Interview

Sub-question 2: What AI tools do exist for a Designers and Engineer?	
Sub-question type	Desk & Field
Goal	What is already available and how dan this be used.
Research-strategy	Document analysis - Interview

Sub-question 3: What parts of the designing can be supported or automated by AI?	
Sub-question type	Desk & Field
Goal	Insight in the possibilities of AI to support or automate task for design-engineers
Research-strategy	Document analysis - Interview

Sub-question 4: What opportunities and risks brings AI for a designers and Engineers?	
Sub-question type	Desk & Field
Goal	Get insight in how design-engineers see the opportunities and risks in the use of AI in their work
Research-strategie	Document analysis - Interview

1.4 Quality criteria for the handbook

- **Understandable**
 - Engineers and designers must be able to understand the content without needing AI expertise.
 - Technical terms are explained when necessary.
- **Practically applicable**
 - The handbook provides recommendations and examples that can be directly applied in daily design work.
- **Balanced**
 - Both opportunities and risks of AI are discussed, allowing readers to make informed decisions.
- **Action-oriented**
 - Readers understand what actions they can take to start using AI in their work.
- **Up-to-date**
 - The handbook includes current AI tools, technologies, and developments.
- **Well-structured**

- Information is logically organized and easy to navigate.
- **Easy to use**
 - Readers can quickly find the information they need through clear headings, page numbers, and a table of contents.
- **Evidence-based**
 - Recommendations are supported by research findings and interview results.

2. AEL – Product requirements document

2.1 Problem and users

2.1.1 Problem

Researching AI applications for designers and engineers is time-consuming. Relevant information is spread across academic papers, company websites, AI tool documentation, industry reports, and interview transcripts. Much of the available information for Dutch SMEs is published in Dutch, while the handbook must be written in English.

The research team needs a platform that can efficiently collect, translate, analyze, and organize information into handbook pages.

2.1.2 Primary Users

- Student research team members
- Non-technical researchers who need to gather and analyze sources

2.1.3 Secondary Users

- SME owners
- Design engineers

These users will ultimately read the handbook to understand how AI can improve their design pipeline.

2.2 What must the user do

The platform should support the complete workflow from source collection to handbook publication.

1. Collect Sources	• Upload documents • Add website URLs • Add interview transcripts • Import Dutch and English sources
2. Analyse Sources	• Summarize documents • Extract key findings • Identify opportunities and risks • Detect information relevant to the research question
3. Translate Content	• Translate Dutch sources into English • Preserve technical terminology • Show original text alongside translations
4. Generate Handbook Draft	• Create handbook page drafts • Generate APA references • Suggest figures and examples
5. Human Review	• Edit generated prompt • Verify sources • Approve final handbook pages
6. Publish	• Export handbook pages

2.3 Output Qualities

A handbook page produced by the platform should:

Output quality	Description
1. Be structured consistently	Each page follows a standard format, including: • Title • Introduction • Main findings • Practical examples • Recommendations • References
2. Be written in clear English	The content on the page is written in professional, easy-to-understand English suitable for engineers, designers, and SME owners.
3. Be traceable to sources	Every key claim, recommendation, and finding is linked to a supporting source or interview.
4. Maintain a consistent style	Pages use the same tone, terminology, formatting, and citation style.
5. include citations	Sources are referenced according to APA 7th edition.

2.4 Known constraints

Requirements for literature / sources

- Literature is no more than 4 years old.
- Sources are academic or professional (no Wikipedia or blogs without an author).
- Language: Dutch and/or English.
- Sources originate from recognized publishers, professional journals, government bodies, or reputable knowledge organizations.

Agentic CLI

- The platform must be operated through an agentic CLI instead of custom-coded software.

Existing Infrastructure

- The project must reuse existing wiki and website infrastructure.

2.5 Source material

Many useful sources for Dutch SMEs are written in Dutch, including:

- CBS reports
- Chamber of Commerce publications
- Sector organizations
- Regional news articles
- Interview transcripts

2.6 Out of scope (for now)

The platform will NOT:

- Build custom AI models
- Replace human decision making
- Conduct interviews itself
- Generate engineering designs
- Integrate directly into CAD software

These features may be considered in future versions.

2.7 Open questions and assumptions